SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-015: Student Records Export to CSV

Field	Details
Change Request ID	CR-015
Change Request Title	Student Records Export to CSV
Student Name	C. Lavasujan
Student ID	24ICT06
Implementation Status	Completed & Verified (as stated in the supplied report)
Date	15 October 2026

1. Change Request Summary

1.1 Business Requirement

CR-015 introduces a secure CSV export capability so authorized administrators can download student records for reporting, offline analysis, and record-keeping. The feature supports both protected backend download endpoints and a client-side export button on the Administrator Dashboard, including export of filtered search results.

1.2 Existing Behaviour

The supplied source identifies CSV export as the capability added by CR-015. It does not describe the prior interface in detail; the change adds the export endpoints, CSV utility, and dashboard controls required for this workflow.

1.3 Requested Change

- Provide GET /api/students/export and GET /api/students/export/csv for authorized administrators using JWT authentication and admin-only role checks.
- Export Student ID, Registration Number, First Name, Last Name, Email, and Phone Number with CSV escaping compliant with RFC 4180.
- Support client-side downloads of either filtered student records or the full list, with success and error feedback.
- Return a valid CSV header row for an empty list and keep export operations read-only against the database.
- Preserve responsive dashboard styling, dark-mode support, and existing student/admin access controls.

2. Implementation

2.1 Implementation Summary

CR-015 was implemented across the backend controller and routes, a frontend CSV utility, the Administrator Dashboard, and dashboard styling. The backend returns a downloadable CSV response and restricts both export endpoints to authenticated administrators. The dashboard utility generates RFC 4180-style CSV content, downloads it as students.csv, and displays the number of records exported.

2.2 Affected Files

File	Change / Role	Purpose
backend/src/controllers/studentController.js	Implemented export handler	exportStudentsCsv(req, res) selects student records and returns escaped CSV with download headers.
backend/src/routes/studentRoutes.js	Routes added	Adds GET /api/students/export and GET /api/students/export/csv before parameterized ID routes; applies authMiddleware and adminOnly.
frontend/src/utils/csvExport.js	CSV export utility	Escapes values, generates standardized CSV text, triggers Blob download, and reports export statistics.
frontend/src/pages/DashboardPage.jsx	Dashboard integrated	Adds Export CSV beside + Add Student, exports filtered or all records, and shows result/error feedback.
frontend/src/styles/dashboard.css	Styling updated	Styles the export button and feedback banner for hover, disabled, responsive, and dark-mode states.
backend/src/scripts/testCr015Export.js	Automated test script	Supports the 15 authentication, response, data-integrity, and empty-list verification cases listed in the source.

2.3 Technical Changes

Backend controller: exportStudentsCsv(req, res) exports six fields with clear column headers. Values containing commas, double quotes, or newline characters are escaped according to RFC 4180. Responses use Content-Type: text/csv; charset=utf-8 and Content-Disposition: attachment; filename="students.csv". An empty list still returns the valid header row.

Routing and security: GET /api/students/export and GET /api/students/export/csv use authMiddleware followed by adminOnly. The export paths are registered before parameterized /api/students/:id routes to avoid route collisions.

Frontend utility: escapeCsvValue, generateStudentsCsv, downloadCsv, and exportStudentsToCsv perform escaping, standard header generation, Blob download, and reporting of success, count, and filename.

Dashboard and styling: the Export CSV button is next to + Add Student, is disabled while data is loading, respects active filters, and displays a success banner with the exported record count and destination filename. Error feedback, responsive sizing, and dark-mode styling are included.

Database safety: the backend export uses SELECT queries only; the supplied tests report that row counts and student record hashes were unchanged by export.

2.4 Implementation Flow

1. An administrator signs in and obtains a valid JWT, then opens the Administrator Dashboard.
2. The administrator selects Export CSV; if a search filter is active, the filtered records are exported, otherwise the full student list is used.
3. The frontend creates CSV content with standardized headers and escaped field values, then triggers a browser download.
4. For backend downloads, an authenticated admin request to either export endpoint returns students.csv with the CSV content type and attachment disposition.
5. The dashboard displays export success with the number of records and filename, or a message if the export/download fails.

3. Testing & Verification

3.1 Testing Approach

The supplied CR-015 report states that an automated test suite covered 15 functional, security, response-format, data-integrity, and empty-list cases. Browser verification covered dashboard button visibility, role-based access control, responsive/light-dark presentation, and the production build.

Environment / Build: Development dashboard/backend; production build executed with npm run build. The source reports zero lint, syntax, or bundling errors.

3.2 Test Cases

Test ID	Test Case	Result / Verification	Status
TC-01	Admin authentication	Administrator logs in and receives a JWT.	Pass
TC-02	Student authentication	Student logs in and receives a JWT for authorization testing.	Pass
TC-03	Request without token	Unauthenticated export request is rejected with HTTP 401.	Pass
TC-04	Student role access	Student token is rejected with HTTP 403.	Pass
TC-05	Admin export response	GET /api/students/export returns HTTP 200 for an administrator.	Pass
TC-06	Content-Type header	Response contains text/csv; charset=utf-8.	Pass
TC-07	Content-Disposition header	Response specifies attachment; filename="students.csv".	Pass
TC-08	Non-empty response	Response includes the CSV header and content.	Pass
TC-09	CSV headers	Headers match Student ID, Registration Number, First Name, Last Name, Email, Phone Number.	Pass
TC-10	Row-count parity	Exported data-row count matches the database count.	Pass
TC-11	Field count	Each exported record contains exactly six fields.	Pass
TC-12	Data integrity	Exported values match the database fields.	Pass
TC-13	Database immutability	Database row count is unchanged after export.	Pass
TC-14	Database record hash	Student rows are identical before and after export.	Pass
TC-15	Empty-list safety	Empty input produces valid CSV headers without error.	Pass

3.3 Test Summary

Test Result	Count / Status
Total test cases	15
Passed	15
Failed	0
Pass rate	100%
Production build	npm run build completed with 0 lint, syntax, or bundling errors.
Overall result	Pass (as stated in the supplied source).

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Only authenticated administrators can use the backend export endpoints.	JWT authentication and admin-only route checks; missing token 401 and student role 403.	Met
AC-002	CSV export provides standardized headers and the six specified student fields.	Header-format, field-count, and data-integrity tests.	Met
AC-003	CSV output escapes commas, quotes, and newline characters according to RFC 4180.	CSV utility/controller escaping logic described in the source.	Met
AC-004	Export downloads with the expected CSV content type and filename.	HTTP response header checks.	Met
AC-005	Export does not modify student records or database metadata.	Row-count and record-hash checks before and after export.	Met
AC-006	Empty input still produces valid CSV headers.	Empty-list safety test.	Met
AC-007	Dashboard can export filtered results or all records and shows feedback.	Browser verification and dashboard integration described in source.	Met
AC-008	Student users do not see the export control and direct export is denied.	Student-dashboard check and HTTP 403 verification.	Met
AC-009	Production build completes without lint, syntax, or bundling errors.	npm run build result reported in source.	Met

5. Requirements Traceability Matrix

Requirement Area	Implementation Details	Status
Backend export	exportStudentsCsv returns six student fields as a downloadable CSV response.	Met
Authentication and authorization	Both backend export routes use authMiddleware and adminOnly.	Met
CSV compatibility	Values with commas, double quotes, and line breaks are escaped; empty lists return headers.	Met
Data integrity	Export uses SELECT queries; tests report unchanged row counts and record hashes.	Met
Dashboard integration	Export CSV button handles filtered/full lists, loading state, and visible feedback.	Met
UI styling and accessibility of states	Responsive button and success banner styling supports light/dark mode and disabled/error states.	Met

6. Evidence

Replace the placeholders below with screenshots captured from the actual application and terminal. Use test data where possible, and do not expose JWT tokens, credentials, or unrelated personal information.

Evidence 01 - Administrator Dashboard with Export CSV Button

Related criteria: AC-007

Show the Administrator Dashboard with Export CSV beside + Add Student. Include a filtered view if demonstrating filtered export.

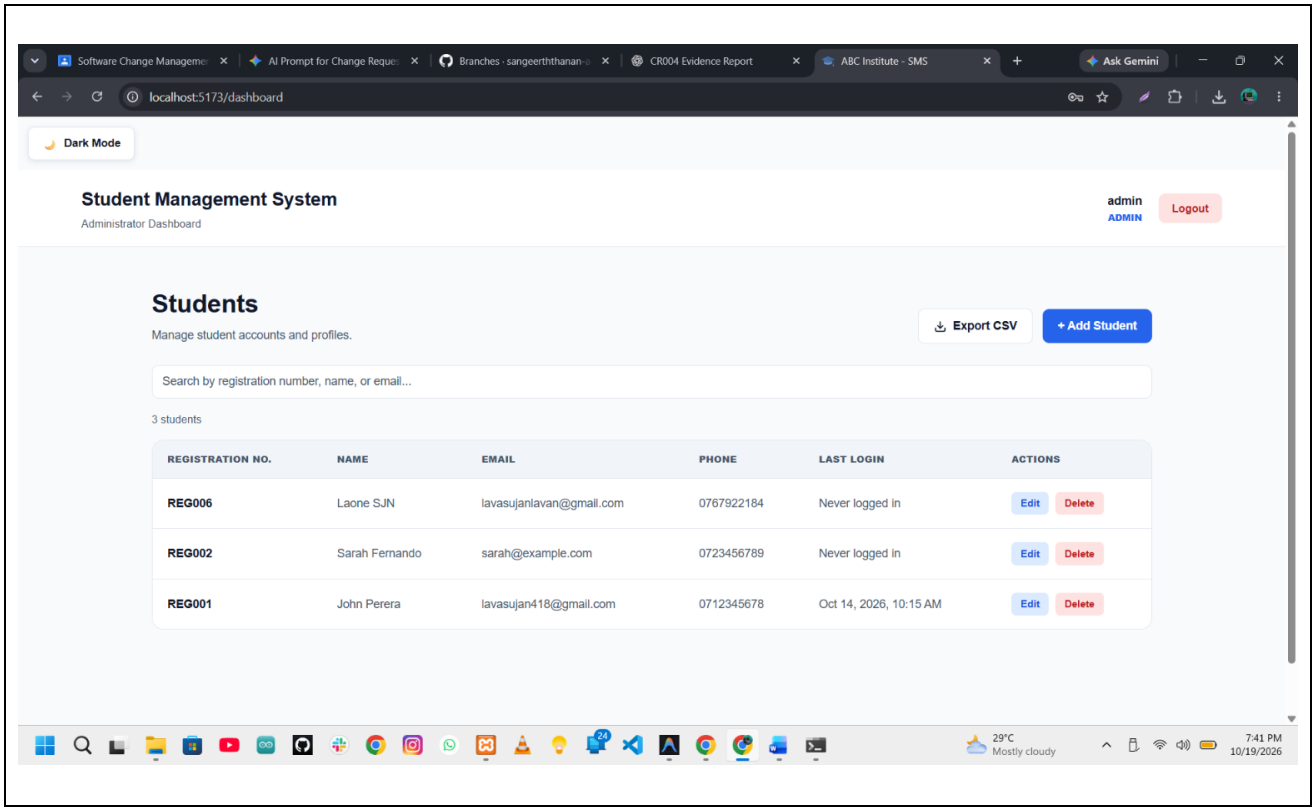


Figure 1: Administrator Dashboard with Export CSV Button.

Evidence 02 - Downloaded CSV File and Column Headers

Related criteria: AC-002, AC-003, AC-004

Show the downloaded students.csv file with the six expected column headers and representative rows. Avoid unnecessarily exposing personal student data.

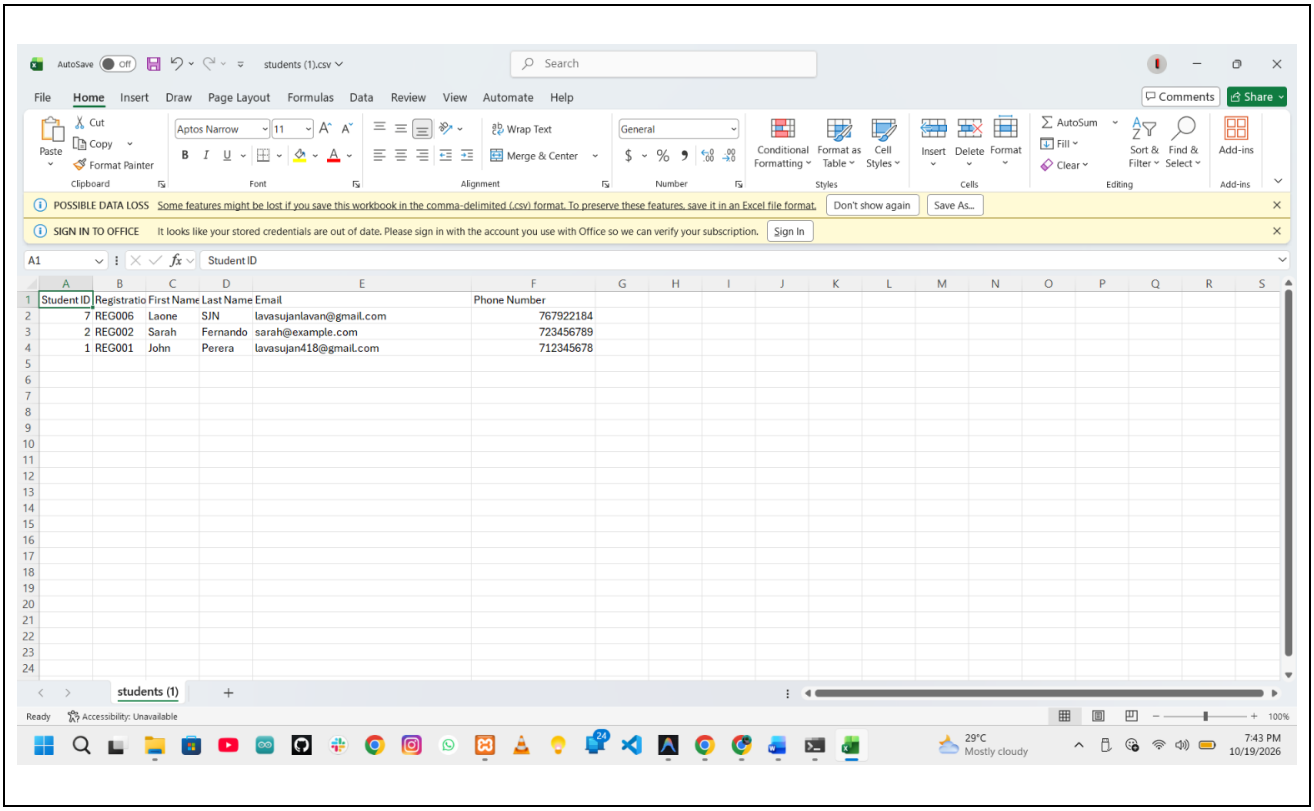


Figure 2: Downloaded CSV File and Column Headers.

Evidence 03 - Role-Based Access Control

Related criteria: AC-001, AC-008

Show that a student account has no Export CSV control and/or an API client response of HTTP 403 when the student token attempts the export route. Do not show the token value.

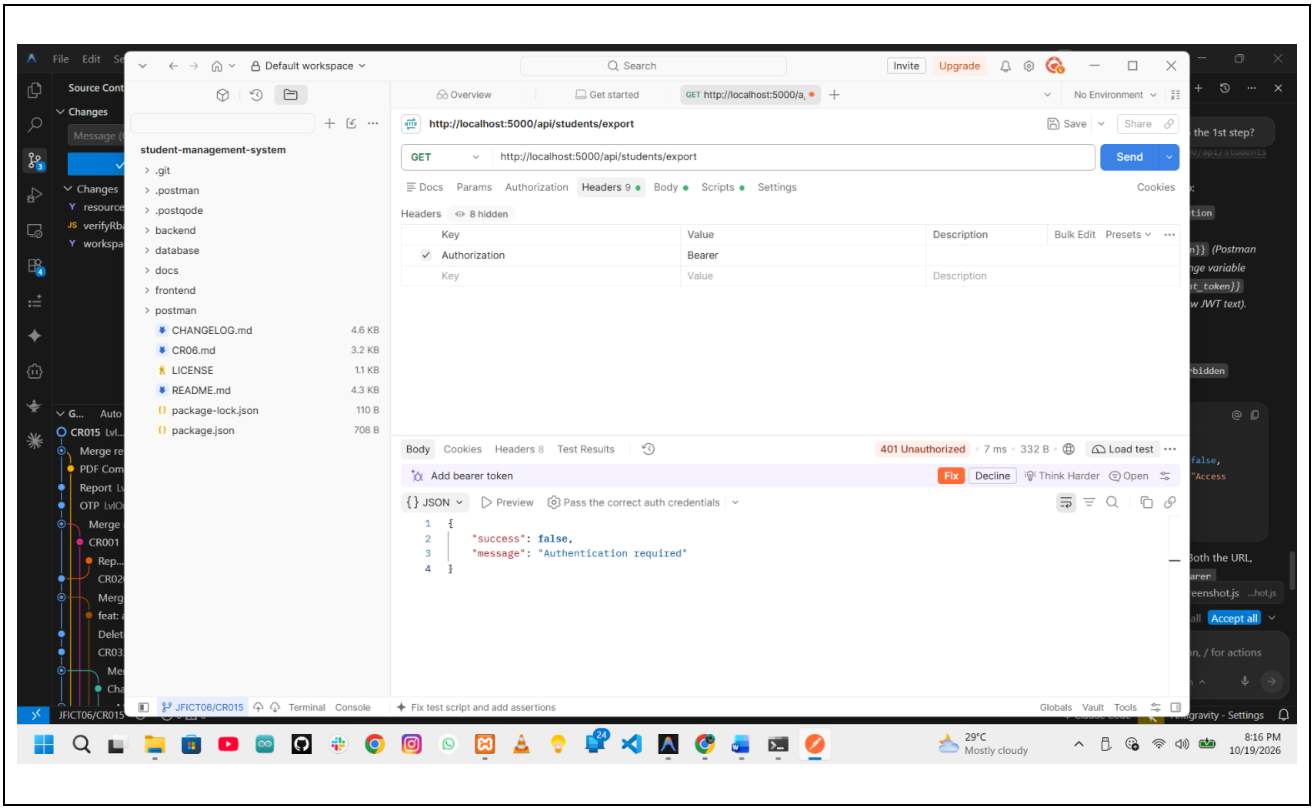


Figure 3: Role-Based Access Control.

Evidence 04 - Automated Test Suite and Production Build

Related criteria: AC-001 to AC-006, AC-009

Capture terminal output from backend/src/scripts/testCr015Export.js showing the actual passing checks and npm run build completion. Do not fabricate output if your local execution differs from the supplied report.

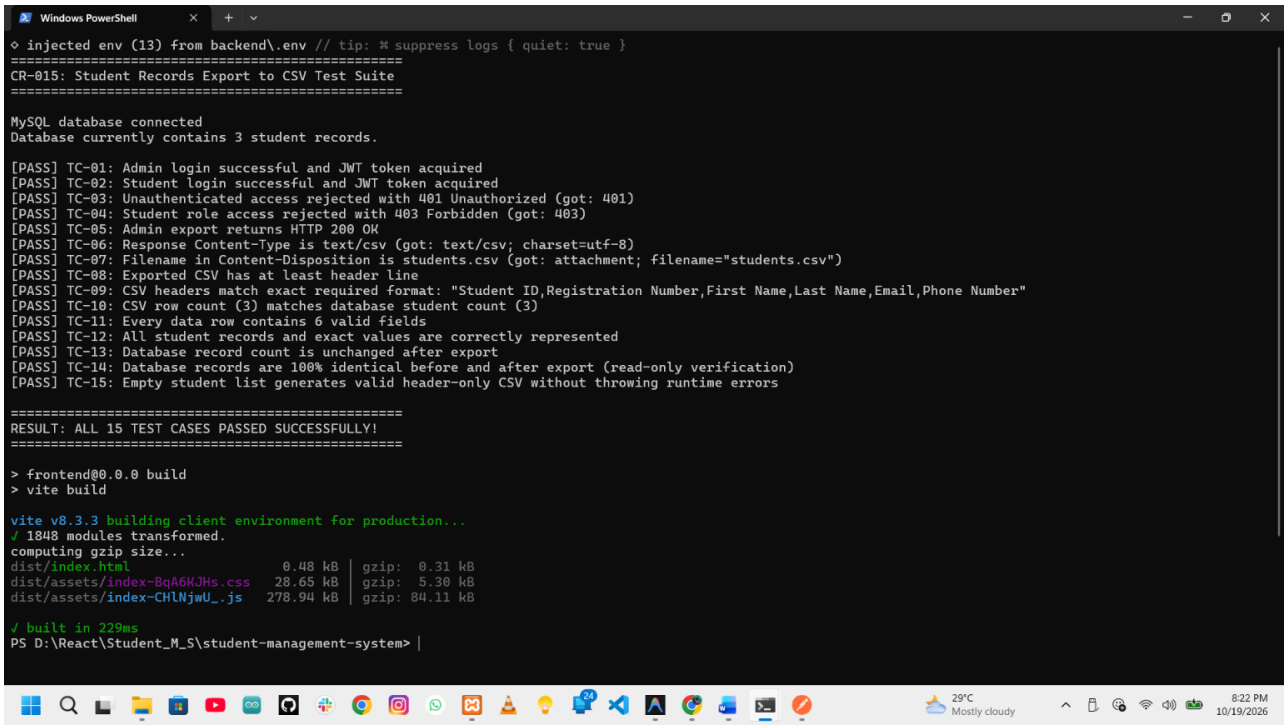


Figure 4: Automated Test Suite and Production Build.

7. Final Result

Item	Result
Requirements	Met (as documented in the supplied report).
Implementation	Completed: backend endpoints, frontend utility, dashboard integration, and styling.
Testing	15 / 15 tests passed; 100% pass rate reported.
Security Controls	JWT authentication and admin-only access reported verified; student export access denied.
Data Integrity	Read-only SELECT behaviour; row count and record hashes reported unchanged.
Production Build	npm run build completed with zero reported lint, syntax, or bundling errors.
Final Status	Completed & Verified (as stated in the supplied source).

7.1 Final Summary

CR-015 adds a CSV export feature to the Student Management System. Administrators can download student records from the dashboard or protected backend endpoints. The export includes six standardized fields, handles CSV-special characters, supports filtered lists and empty results, and uses read-only database access. The supplied source reports all 15 automated tests passed, student-role access is denied, and the production build completed successfully.

7.2 Issues / Limitations

No issues or limitations are identified in the supplied CR-015 report. The evidence screenshots should be captured from the actual running application and local test output before final submission.

Student Name: C. Lavasujan

Student ID: 24ICT06

Date: 15 October 2026

Signature: _____